

ARUNDHATI SENGUPTA

Computational Linguist — LLM Data, Annotation & Evaluation
arundhatisengupta2@gmail.com | 872-202-2007 | [Linkedin](#) | [GitHub](#)

SUMMARY

Linguist with 10+ years of experience in language data and technology across speech and text, spanning many languages, scripts, and dialects. My foundation is language. I hold Ph.D. in Computational Linguistics (CUNY Graduate Center); my dissertation includes the first systematic evaluation of LLMs on Bengali grapheme-to-phoneme conversion. Experienced in end-to-end data work (collection, annotation guidelines, annotator QC, and model evaluation) using Python, SQL, and finite-state rules. Published at ACL/SIGMORPHON.

CORE SKILLS

Programming & Data: Python (PyTorch, Hugging Face Transformers, scikit-learn, Pynini, TensorFlow, Fairseq), SQL, R, UNIX, Git, Jupyter, spreadsheets (Excel/Sheets)

Rule Encoding: Regular expressions and finite-state (WFST) rewrite rules for pattern extraction, text normalization, and rule-based classifier encoding

LLMs: Prompt engineering (zero-/few-shot design), benchmarking LLMs (Gemini, Claude, GPT-4o) via API against task baselines, systematic LLM-output error analysis

Data & Annotation: Annotation guideline authoring, annotator training and QC, inter-annotator agreement, data collection protocols, large-scale multilingual corpus curation, A/B testing, reproducible study design

Linguistics: Syntax, semantics, pragmatics, sociolinguistics (dialect and register variation), corpus linguistics, phonetics and phonology, morphology

Languages: English (native), Bengali (native), Hindi (professional); applied analysis across multiple high- and low-resource South Asian languages and scripts

EXPERIENCE

Research Assistant — Language Data & Model Evaluation | RF, CUNY

Aug 2023 – Sep 2025

- Built and quality-controlled large-scale multilingual annotated corpora with native-speaker annotators; **authored annotation guidelines** and adjudicated ambiguous cases so annotators labeled consistently at quality (BSF grant).
- Designed zero-shot and few-shot **prompts** and ran a systematic API-based benchmark of four LLMs (Gemini 2.5 Pro, Claude Opus 4.5, Claude Sonnet 4, GPT-4o) on a structured linguistic annotation task; manually categorized model errors by type and root cause to guide prompt design and model selection.
- Developed grapheme-to-phoneme and diacritization models in **Python/PyTorch** using neural sequence methods; maintained reproducible train/eval splits across languages and scripts.
- Integrated heterogeneous multilingual datasets using **SQL and Python** (PSC-CUNY grant); tuned models with Weights & Biases and achieved a **10% relative error reduction** over baseline through evaluation-driven refinement.

Linguistic Data Scientist | Central Institute of Indian Languages

Oct 2005 – Oct 2012

- Led speech and text corpus projects across multiple Indian languages and dialects, designing **data collection protocols, annotation guidelines, and evaluation metrics** for large-scale multilingual, multi-script language resources.
- Ran A/B testing and systematic error analysis to assess data and resource quality at scale, with close attention to dialectal and register variation.
- Acted as liaison between linguists, annotators, and engineers, translating linguistic findings into concrete product and pipeline improvements.

Speech Data Evaluator | Google Inc.

Nov 2012 – Sep 2013

- Assessed acoustic and phonemic model performance on large-scale speech datasets using quantitative metrics; designed phonemic inventories and reported evaluation findings to cross-functional engineering teams to improve models and training data.

Freelance Linguistic QA Reviewer | Lionbridge Technologies, Inc.

Sep 2013 – Aug 2016

- Performed linguistic quality assurance on multilingual software output, providing structured, actionable feedback to engineering and localization teams.

SELECTED PROJECTS

- G2P Error Analysis Tool (open source)** — WFST/rule-based QA tool automating quality control and systematic error reporting on model output.
- Personality Trait Prediction from Speech (M.A. thesis)** — statistical and machine learning modeling of acoustic-prosodic features across gender and L1 groups.

EDUCATION

Ph.D. in Linguistics, CUNY The Graduate Center — Dissertation: *A study of Bengali Grapheme-to-Phoneme Conversion*

Aug 2019 – Sep 2026

M.A. in Linguistics, CUNY The Graduate Center

2016 – 2018

M.A. in Linguistics, University of Calcutta, India | **B.A. in Linguistics**, University of Calcutta, India

2000 – 2005

SELECTED PUBLICATIONS

- Sengupta, A. et al. (2021). Results of the Second SIGMORPHON Shared Task on Multilingual Grapheme-to-Phoneme Conversion. Association for Computational Linguistics.
- Sengupta, A. (2016). Voice Onset Time and Burst Duration of Bangla Labial Stop Consonants in CV Context. *Language in India*, 16(3), 37–45.
- Editor, Longman-CIIL Basic English-English-Bengali Dictionary (Pearson/Longman); Editor, Proceedings of Multilingual Speech Resources, CIIL, 2010.